

CSE 2215: Data Structure and Algorithms-I

1. Using the Iterative Binary/Linear Search Algorithms find the key x from the following data where ID=last digit of your student ID. Show step by step iterations where x=ID.

10 40 20 8 99 ID 15 17

2. Using the Recursive Binary/Linear Search Algorithms draw a recursion tree for each of the searching techniques to find the key x from the following data where ID=last digit of your student ID.

10 40 20 8 99 ID 15 17

3. Find the element comparisons for each of the searching techniques of the Ques. 1

4. Find an equation for the column-wise memory allocation of 2D array.

5. Find the location of A[65] for the following data

int A[100]

Assume, loc(A[0])= (AE92F6)_H

6. Find the location of A[15] [20] for the following data

int A[50][100]

Assume, loc(A[0][0])= (AE92F6)_H

Assume column wise memory allocation

7. Show the structure memory allocation for the following with structure padding. Assume 32-bit and 64-bit computer. Also assume that starting location of variable a is 1000.

```
struct sample{
    char a;
    char b;
    char c;
    double d;
}e;
```

How many bytes memory is needed for e if structure padding is used or not used?

8. For the following linear linked list show memory allocation. Assume the locations 2005, 2003, 2010, 2008, 2007, 2009 are free for allocation. Also assume starting node name is head. Show the memory mapping from the location 2001 to 2010.

20->90->50->40->55->38->NULL

9. Discuss the best, worst, and average cases element comparisons of linear and binary searches.